

Name: _____

SOLUTIONS

Date: _____



Year 8 Mathematics

Test 3, 2015

Topics – Volume, Fractions, Decimals, Percentages

 64

 = ____ %

Total Time: 4 minutes
 Total Reading: 4 minutes
 Total Working: 55 minutes
 Weighting: % of the year.

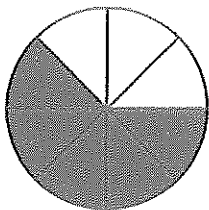
CALCULATOR FREE

Time:	27 minutes	Marks for Section 1:	32
Reading:	2 minutes	Equipment Allowed:	Nil
Working:	25 minutes		

1. [20 marks]

MULTIPLE CHOICE: Circle the answer you think is most correct. Each question is worth 1 mark

a) What fraction of the circle is shaded?

A $\frac{3}{5}$ B $\frac{3}{8}$ C $\frac{5}{8}$ D $\frac{5}{3}$

b) What decimal fraction is represented by the shaded part of the rectangle?



A 0.2

B 0.4

C 0.5

D 0.6

c) Write $\frac{12}{30}$ in simplest form.

A $\frac{6}{15}$ B $\frac{4}{10}$ C $\frac{2}{5}$ D $\frac{3}{10}$

d) Which of the following is an improper fraction?

A $\frac{6}{7}$

B $\frac{4}{3}$

C $1\frac{2}{9}$

D $\frac{99}{100}$

e) Write $4\frac{2}{5}$ as an improper fraction.

A $\frac{6}{5}$

B $\frac{5}{13}$

C $\frac{20}{5}$

D $\frac{22}{5}$

f) Write $\frac{26}{8}$ as a simplified mixed fraction.

A $3\frac{1}{4}$

B $4\frac{1}{3}$

C $3\frac{2}{8}$

D $\frac{13}{4}$

g) Write 0.36 as a simplified fraction.

A $\frac{36}{100}$

B $\frac{18}{50}$

C $\frac{9}{25}$

D $\frac{36}{50}$

h) Change $2\frac{5}{8}$ to a decimal.

A 2.58

B 2.85

C 2.625

D 2.225

i) Change $\frac{5}{6}$ to a decimal.

A 0.56

B 0.83

C 0.53

D 0.83

j) What is the value of ? in $\frac{3}{5} = \frac{?}{20}$

A 4

B 5

C 12

D 18

k) What is half of a half?

A $\frac{1}{2}$

B $\frac{1}{4}$

C $\frac{2}{2}$

D $\frac{1}{8}$

l) Find $\frac{2}{9} + \frac{5}{9}$

A $\frac{7}{9}$

B $\frac{7}{18}$

C $\frac{2}{3}$

D $1\frac{2}{9}$

m) Find $\frac{2}{3} \times \frac{4}{5}$

A $\frac{2}{15}$

B $\frac{6}{15}$

C $\frac{3}{5}$

D $\frac{8}{15}$

n) Work out $2\frac{2}{3} \div \frac{1}{9}$

A $\frac{8}{9}$

B $\frac{3}{8}$

C 8

D 24

o) Work out $\frac{4}{9} - 2$

A $\frac{5}{9}$

B $1\frac{5}{9}$

C $-\frac{5}{9}$

D $-1\frac{5}{9}$

p) $25.073 \div 100 =$

A 0.025 073

B 0.250 73

C 2507.3

D 25 073

q) Estimate an answer to $\frac{4}{11} \times 5.974 - 7\frac{9}{10} \times 0.48 = \frac{1}{3} \times 6 - 8 \times 0.5 = 2 - 4 = -2.$

A 2

B -1

C -2

D -3

r) Which of these would give you the most money?

A $\frac{2}{3}$ of \$690

B $\frac{3}{4}$ of \$500

C $\frac{4}{9}$ of \$972

D $\frac{11}{12}$ of \$480

s) Find $\frac{3}{7}$ of 42.

A 14

B 18

C 21

D 28

t) What is 2% of 3000?

A 0.6

B 6

C 60

D 600

SHORT ANSWER: Show full working out to be awarded maximum marks.

2. [1 mark]

$\frac{2}{5}$ of a 20 ha paddock is no good for planting. How many hectares can be planted?

$$\frac{3}{5} \times \frac{20}{1} = \frac{60}{5} = 12 \text{ ha } \checkmark$$

3. [3 marks: 1, 1, 1]

Calculate each of the following:

a) $\frac{2}{8} + \frac{3}{8} = \frac{5}{8} \checkmark$

b) $\frac{2}{3} \times \frac{2}{5} = \frac{6}{15} = \frac{2}{5} \checkmark$

c) $\frac{4}{7} \div \frac{2}{5} = \frac{4}{7} \times \frac{5}{2} = \frac{20}{14} = \frac{10}{7} = 1\frac{3}{7} \checkmark$

4. [8 marks: 2, 2, 2, 2]

Calculate each of the following. (show full working)

a) $\frac{5}{8} + \frac{7}{12} = \frac{15}{24} + \frac{14}{24} \checkmark$
 $= \frac{29}{24}$
 $= 1\frac{5}{24} \checkmark$

b) $3\frac{3}{5} - 4\frac{1}{10}$
 $= \frac{18}{5} - \frac{41}{10}$
 $= \frac{36}{10} - \frac{41}{10} \checkmark$
 $= \frac{-5}{10} = -\frac{1}{2} \checkmark$

c) $1\frac{1}{20} \times \frac{15}{28}$
 $= \frac{213}{420} \times \frac{153}{284} \checkmark$
 $= \frac{9}{16} \checkmark$

d) $5\frac{1}{3} \div 1\frac{1}{10}$
 $= \frac{16}{3} \div \frac{11}{10}$
 $= \frac{16}{3} \times \frac{10}{11} \checkmark$
 $= \frac{160}{33}$
 $= 4\frac{28}{33} \checkmark$

Name: _____

CALCULATOR ALLOWED

Time:	32 minutes	Marks for Section 2:	32
Reading:	2 minutes	Equipment Allowed:	1 A4 page of notes, Calculator
Working:	30 minutes		

5. [5 marks: 1, 1, 1, 1, 1]
Insert <, =, or > to make true statements.

a) $\frac{5}{6} \boxed{>} \frac{5}{7}$ ✓

b) $\frac{7}{8} \boxed{<} \frac{11}{12}$ ✓

c) $\frac{15}{20} \boxed{=}$ $\frac{21}{28}$ ✓

d) $4.72 \boxed{>} 4.719$ ✓

e) $0.42 \boxed{<} \frac{3}{7}$ ✓

6. [1 mark]
What fraction of an hour is 24 minutes?

$$\frac{24}{60} = \frac{2}{5} \quad \checkmark$$

7. [1 mark]
What percentage of an hour is 15 minutes?

$$\frac{15}{60} \times \frac{100}{1} = \frac{1}{4} \times \frac{100}{1} = 25\% \quad \checkmark$$

8. [2 marks]
Write in order from smallest to largest:

$$1\frac{8}{9}, 0.26, -0.27, -\frac{2}{7}, 2\frac{3}{5}$$

-0.285

$$-\frac{2}{7}, -0.27, 0.26, 1\frac{8}{9}, 2\frac{3}{5} \quad \checkmark\checkmark$$

-1/2 each error

9. [3 marks]

Fill in the missing spaces to make each row full of equivalent values. The first row has been done for you.

Decimal	Percentage	Fraction
0.45	45%	$\frac{9}{20}$
0.3 $\frac{1}{2}$	30%	$\frac{3}{10}$ $\frac{1}{2}$
0.03	3%	$\frac{3}{100}$ $\frac{1}{2}$
0.4 $\frac{1}{2}$	40% $\frac{1}{2}$	$\frac{2}{5}$

10. [2 marks]

If \$200 is increased by 15% and then decreased by 10% how much money remains?

$$\begin{aligned}
 &200 \times \frac{15}{100} = 30 \\
 &\therefore 200 + 30 = 230 \checkmark \\
 &230 \times \frac{10}{100} = 23 \\
 &230 - 23 = \$207 \checkmark
 \end{aligned}$$

11. [3 marks: 1, 1, 1]

Michaela is saving up to buy a unicorn. Her parents offer to give her an extra 10% of any money she saves.

She earns the following amounts: \$415.50, \$1270, \$1820.55; and saves all of it.

a) How much has she saved?

$$\begin{aligned}
 &415.50 + 1270 + 1820.55 \\
 &= \$3506.05 \checkmark
 \end{aligned}$$

b) How much will her parents give her?

$$3506.05 \times \frac{10}{100} = \$350.61 \checkmark$$

c) How much will she have in total to buy a unicorn?

$$\begin{aligned}
 &3506.05 + 350.61 \\
 &= \$3856.66 \checkmark
 \end{aligned}$$

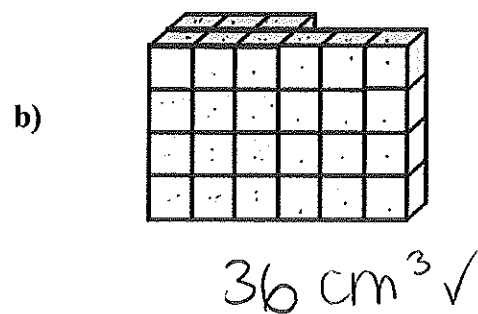
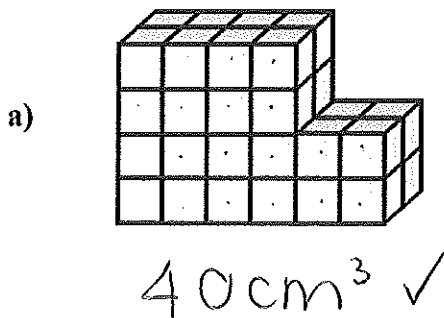
12. [2 marks: $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$]

Complete the following volume conversions:

- a) $2.5 \text{ cm}^3 = \underline{2500} \text{ mm}^3$ $\frac{1}{2}$ b) $0.389 \text{ km}^3 = \underline{389000000} \text{ m}^3$ $\frac{1}{2}$
 c) $18967 \text{ cm}^3 = \underline{0.018967} \text{ m}^3$ $\frac{1}{2}$ d) $47890 \text{ cm}^3 = \underline{47.89} \text{ L}$ $\frac{1}{2}$

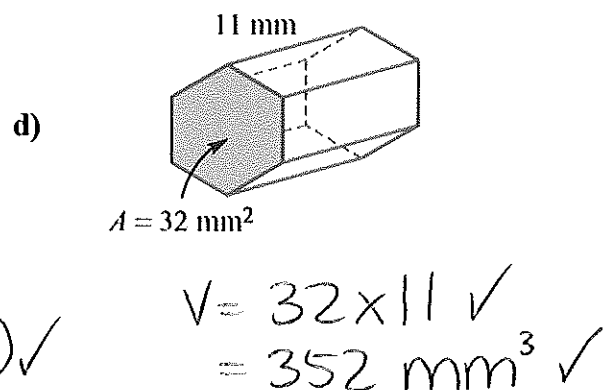
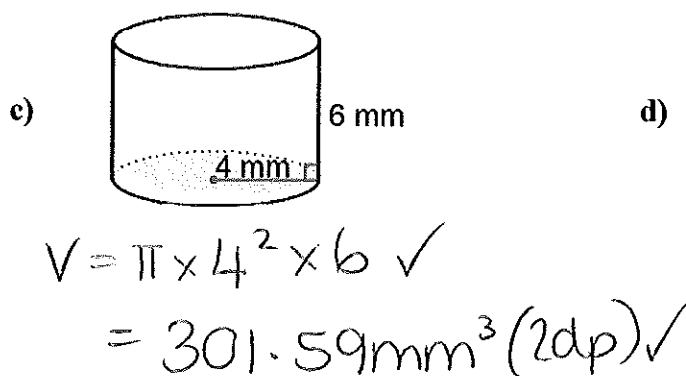
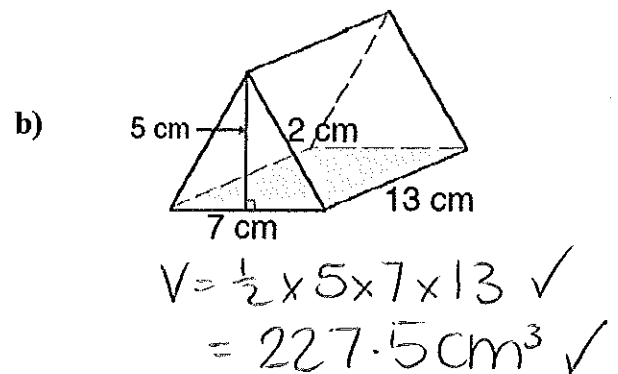
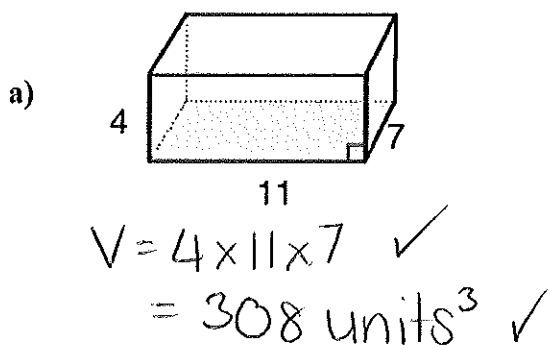
13. [2 marks: 1, 1]

Determine the volume of each of the following shapes by counting cubes. Each cube is one cubic centimetre. Don't forget about the hidden cubes.



14. [8 marks: 2, 2, 2, 2]

Determine the volume of each of the following shapes.

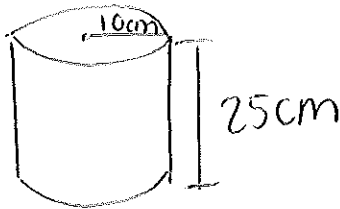


15. [3 marks]

David is filling his petrol container for his lawn mower on a cheap fuel day.

His tank is cylindrical, with a radius of 10 cm, and a height of 25 cm.

If he only needs to fill 80% of his tank with petrol, how much fuel will he purchase? Give your answer in litres.



$$\begin{aligned} V &= \pi \times 10^2 \times 25 \\ &= 7853.98 \text{ cm}^3 \quad \checkmark \\ &= 7853.98 \text{ mL} \\ &= 7.85 \text{ L (2dp)} \quad \checkmark \end{aligned}$$

$$\therefore 7.85 \times \frac{80}{100} = 6.28 \text{ L (2dp)} \quad \checkmark$$